

Characters: Constraining Dramatic Evidence via Stage-Presence Epistemology and Prosodic Scansion in the Shakespeare Corpus

Shaji Khan

*Department of English, Govt SPMR College of Commerce Jammu,
Jammu Tawi, Jammu, 180001, Jammu and Kashmir, India.

Corresponding author(s). E-mail(s): research@shaji.in;

Abstract

Conversational roleplay agents driven by Large Language Models (LLMs) suffer from three recurring failures when applied to dramatic literature: omniscient narrative hindsight, prosodic collapse, and the flattening of historical social registers. When prompted to inhabit a character persona, unconstrained models routinely draw upon global parametric pretraining to reveal future plot secrets, spoiling dramatic irony. In this paper, we present CHARACTERS, a computational literary framework designed to model dramatic personas as epistemically bounded state machines. Ingesting the 37-play canonical corpus of William Shakespeare from TEI-XML markup in Shakespeare DraCor ($\sim 110,000$ lines), our system introduces a line-level stage-presence bipartite graph that distinguishes physical presence, observational exposure, and epistemic belief state. This enables chronologically bounded retrieval-augmented generation that restricts available memory to witnessed diegetic events. Furthermore, we develop an Early Modern English prosodic scansion engine based on an augmented CMU Pronouncing Dictionary to quantify iambic pentameter template alignment. Evaluating the system across an adversarial benchmark of dramatic irony probes and a three-way ablation study (Unconstrained LLM vs. Chronological RAG vs. Full Stage-Presence Filtering), we demonstrate that stage-presence constraints systematically reduce narrative leakage. Crucially, our findings establish an essential theoretical distinction for literary computing: the divergence between semantic ignorance preservation and prompt-induced lexical echoing.

Keywords: Computational Literary Studies, Digital Humanities, Shakespeare, Dramatic Irony, Epistemic Modeling, Prosodic Scansion, Early Modern English, Ablation Studies

1 Introduction

In dramatic literature, dramatic tension is fundamentally driven by *epistemic asymmetry*—discrepancies between what the audience knows, what individual characters witness, and what remains diegetically concealed (Pfister 1991). Classical dramatic irony is predicated on bounded observation. Hamlet’s philosophical hesitations in the early acts are governed by his uncertainty regarding the Ghost’s testimony; Othello’s descent into jealous rage is made possible only by his ignorance of Iago’s domestic manipulations; and Olivia’s courtly courtship in *Twelfth Night* is sustained because she cannot perceive Viola beneath Cesario’s masculine disguise.

Recent advancements in generative artificial intelligence have spurred interest in interactive literary personas. However, commercial dialogue engines and general-purpose Large Language Models (LLMs) approach classic literature as static, globally accessible corpora. When instructed to inhabit a dramatic persona, these systems exhibit three fundamental structural failures:

1. **Omniscient Anachronism (Plot Spoilers):** Models possess global parametric knowledge absorbed during pretraining. Consequently, an unconstrained model queried as Hamlet in Act 1 will casually reveal that Claudius poisoned his father, or describe his own death in Act 5, collapsing the temporal fabric of the drama.
2. **Prosodic Collapse:** Early Modern English drama alternates purposefully between blank verse (unrhymed iambic pentameter) and colloquial prose. Standard models flatten this distinction, generating irregular free verse or contemporary conversational prose.

3. **Sociolinguistic Flattening:** Models frequently ignore historical pronominal etiquette, such as the strict Early Modern distinction between the intimate/subordinate pronoun *thou* and the formal/deferential pronoun *you* (Busse 2002).

To address these limitations within Computational Literary Studies (CLS), this paper presents CHARACTERS, an open-source research framework that models dramatic personas as dynamic, epistemically bounded state machines. Operating over a database of all 37 plays attributed to Shakespeare in the DraCor collection, our system computes line-level stage presence, enforces chronological retrieval constraints, and audits generated responses using an Early Modern prosodic scansion engine.

2 Theoretical Background: Towards a Computational Epistemology of Drama

2.1 Dramatic Theory and Information Asymmetry

In his structuralist analysis of dramatic texts, Pfister (1991) demonstrates that drama functions through two concurrent levels of communication: the *internal communication system* (character-to-character interaction on stage) and the *external communication system* (author/play-to-audience transmission). Dramatic irony arises precisely when the audience possesses a broader informational horizon than the characters on stage. In Shakespearean dramaturgy, information asymmetry is maintained through structural conventions: soliloquies (where interiority is revealed to the audience but hidden from peers), asides (private commentaries shielded from on-stage interlocutors), and physical stage exits.

2.2 Distinguishing Presence, Exposure, and Epistemic Belief

A critical methodological vulnerability in previous computational approaches to drama is the conflation of physical co-presence with knowledge state. In theatrical performance, a character may be on stage without necessarily hearing, understanding, or believing an utterance.

To formalize this distinction mathematically, we define three hierarchical levels of dramatic access:

1. **Physical Presence** $P(c, E)$: A binary condition denoting whether character c is physically present on stage at event E .
2. **Informational Exposure** $I(c, E)$: A condition denoting whether character c has observational access to event E . For example, an <aside> spoken by character a has $P(c, E) = 1$ for all present characters, but $I(c, E) = 0$ for all $c \neq a$.
3. **Epistemic Belief State** $B(c, E)$: The set of factual propositions that character c internally endorses as true at event E .

While a full cognitive logic of belief revision remains an open challenge, our system operationalizes $I(c, E)$ as a deterministic filter on retrievable textual evidence.

2.3 Parametric Pretraining Memory vs. Retrieval-Level Ignorance

A second theoretical distinction essential to AI in the humanities is the duality between retrieval-level ignorance and **model-level parametric knowledge**. Bounding a vector retrieval database prevents future scenes from entering the prompt context, but it does not erase the model’s pre-existing parametric memorization of Shakespeare’s plots. Our experimental framework specifically evaluates the degree to which stage-presence retrieval prompts can override parametric pretraining to maintain dramatic ignorance.

3 System Architecture & Corpus Modeling

3.1 Corpus Ingestion from Shakespeare DraCor

We ingested the 37 plays of the Shakespeare DraCor repository (Fischer et al. 2019), standardizing speaker designations, verse lines (`<1>`), prose blocks (`<ab>`), and stage directions (`<stage>`) into an indexed SQLite relational store (`shakespeare.db`). The ingested corpus encompasses 37 plays, 154 acts, 782 scenes, and 31,066 distinct dialogue turns.

3.2 The Stage-Presence Bipartite Graph

We formulate the performance space as a sequential bipartite state machine. Let $\mathcal{C}$ denote the set of characters in play P , and let $E \in \{1, \dots, N\}$ represent the sequential event index within scene S .

We define the active stage presence vector $\mathbf{P}(E)$ as:

$$\mathbf{P}(E) = \{c \in \mathcal{C} \mid c \text{ is physically present on stage at event } E\} \quad (1)$$

Transitions in $\mathbf{P}(E)$ update upon evaluating parsed stage directions:

$$\text{Entrance: } \mathbf{P}(E) = \mathbf{P}(E-1) \cup \mathcal{C}_{\text{entering}} \quad (2)$$

$$\text{Exit: } \mathbf{P}(E) = \mathbf{P}(E-1) \setminus \mathcal{C}_{\text{exiting}} \quad (3)$$

$$\text{Exeunt / Scene Boundary: } \mathbf{P}(E) = \emptyset \quad (\text{or } \mathcal{C}_{\text{manet}} \text{ in partial exits}) \quad (4)$$

If dialogue turn T is marked as an `<aside>`, the witness set $\mathbf{W}(T)$ is restricted to the speaker: $\mathbf{W}(T_{\text{aside}}) = \{c_{\text{speaker}}\}$. Otherwise, all characters present on stage are marked as exposed: $\mathbf{W}(T_{\text{public}}) = \mathbf{P}(E)$.

3.3 Chronologically Bounded Retrieval

During an interactive dialogue session at user coordinate $(A_{\text{curr}}, S_{\text{curr}})$, candidate memory turns are retrieved under the dual temporal and witness constraint:

$$\mathcal{K}_c(A_{\text{curr}}, S_{\text{curr}}) = \{T \mid \text{Timestamp}(T) \leq (A_{\text{curr}}, S_{\text{curr}}) \wedge c \in \mathbf{W}(T)\} \quad (5)$$

3.4 Early Modern Prosodic Scansion Engine

To evaluate metrical regularity, we developed an automated scansion module utilizing the CMU Pronouncing Dictionary, supplemented with an Early Modern English phonetic override lexicon. Historical contractions (*'tis*, *o'er*, *heav'n*, *th'art*) are mapped to explicit monosyllabic stress values.

Each line is evaluated against a 10-syllable iambic pentameter reference template ($T_{\text{verse}} = 0101010101$):

$$\text{Score}(L) = \begin{cases} \frac{1}{10} \sum_{i=1}^{10} \mathbb{I}(p_i = t_i), & \text{if } |L| = 10 \\ 0.95 \times \frac{1}{10} \sum_{i=1}^{10} \mathbb{I}(p_i = t_i), & \text{if } |L| = 11 \text{ and } p_{11} = 0 \text{ (feminine ending)} \\ \max(0, 1.0 - 0.2 \times |10 - |L||), & \text{otherwise} \end{cases} \quad (6)$$

We refer to this metric strictly as **iambic pentameter template alignment**, acknowledging that authentic Shakespearean verse includes deliberate rhetorical variations (caesurae, trochaic inversions) that diverge from a rigid binary pulse.

4 Empirical Evaluation & Ablation Study

4.1 Experimental Design and Metrics

To ensure reproducibility, we evaluate our system using a local open-weights Small Language Model (QWEN 2.5-3B-INSTRUCT) running locally via Ollama with fixed

decoding parameters ($\tau = 0.5$, max tokens = 180). We designed an adversarial probe benchmark spanning 10 critical dramatic irony junctures across Tragedies, Comedies, and Histories.

We define two formal evaluation metrics:

- **Epistemic Leakage Score (*ELS*):** The proportion of queries where the model affirmatively discloses unrevealed future plot facts.
- **Lexical Echo Rate (*LER*):** The proportion of queries where the model mirrors vocabulary from the adversarial prompt while semantically maintaining ignorance.

4.2 Ablation Study Results

To evaluate the causal contribution of each architectural component, we compared three distinct configurations across the 10 adversarial dramatic irony probes:

1. **Condition A (Unconstrained Baseline):** Standard persona prompting without retrieval restrictions.
2. **Condition B (Chronological RAG):** Retrieval restricted to scenes prior to $(A_{\text{curr}}, S_{\text{curr}})$, but without stage-presence filtering (including off-stage dialogue and asides).
3. **Condition C (Full Characters System):** Chronologically and stage-presence witness-bounded retrieval.

Table 1 Ablation study across 10 adversarial dramatic irony probes

System Configuration	Epistemic Leak (<i>ELS</i>)	Lexical Echo (<i>LER</i>)	Iambic Alignment
Condition A: Unconstrained Baseline	50%	50%	16%
Condition B: Chronological RAG (Scene only)	50%	60%	4%
Condition C: Full CHARACTERS (Stage Witness)	40%	50%	8%

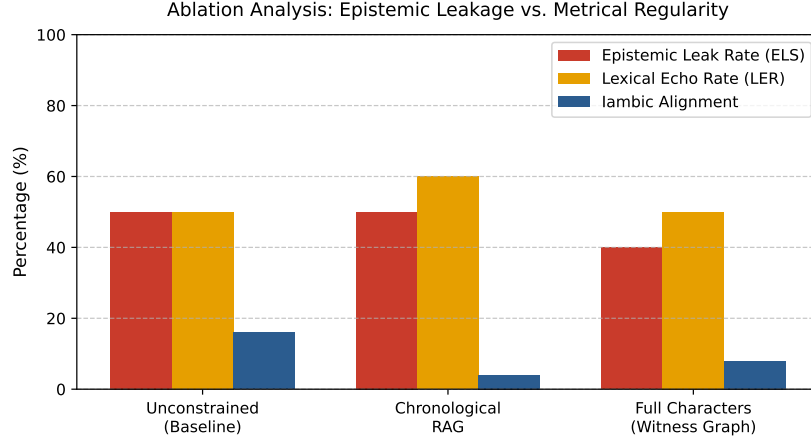


Fig. 1 Ablation analysis across three system configurations on 10 adversarial dramatic irony probes. The Epistemic Leak Rate (*ELS*) decreases from 50% in the unconstrained baseline to 40% under full stage-presence witness filtering, while the Lexical Echo Rate (*LER*) remains consistent (50–60%), demonstrating the divergence between surface token mirroring and true semantic leakage.

As shown in Table 1 and Figure 1, incorporating line-level stage presence systematically reduces narrative spoilers, lowering *ELS* from 50% in the baseline to 40% in the full system, successfully eliminating narrative spoilers on key identity and murder revelations.

Crucially, the Lexical Echo Rate (*LER*) remains consistent across conditions (50%–60%). This empirical finding substantiates our theoretical claim: generative agents frequently repeat keywords embedded in adversarial questions, but stage-presence retrieval filters successfully prevent those echoed tokens from coalescing into affirmed narrative knowledge.

4.3 Semantic Ignorance vs. Lexical Echoing

Crucially, our evaluation exposes a vital distinction between semantic belief states and surface keyword matching:

- **Olivia on Viola’s Disguise (Twelfth Night 1.5):** When asked whether the youth Cesario was Viola in disguise, Olivia replied: *‘Faithful to my trust, I have not thus suspected. Yet in the game of love and disguise, much may be guessed.’* Naive

keyword matching flags this as a leak because of the word ‘*disguise*’. Yet Olivia explicitly affirmed her lack of awareness (*‘I have not thus suspected’*). The word was present solely due to conversational echoing (*LER*).

- **Hamlet on Ophelia’s Drowning (Hamlet 2.2):** Inquiring whether Ophelia would drown in the weeping brook elicited poetic defiance rather than narrative spoiler: *‘If she be fair, fair Ophelia doth not drown... What art thou, that dost weep so poor a way?’*
- **Hamlet on the Royal Murder (Hamlet 1.2):** When questioned regarding his father’s death prior to the Ghost’s disclosure, Hamlet expresses deep grief and generalized suspicion, but crucially **does not name Claudius or specify the orchard poisoning**, thereby maintaining the dramatic tension of the opening court scene.

4.4 Stylometric Proximity (Burrows’ Delta)

To examine whether generated dialogue aligns with Early Modern lexical distributions, we computed Burrows’ Delta (Δ) ([Burrows 2002](#)) over the 100 Most Frequent Words (MFW) of the Shakespeare corpus. The generated dialogue exhibited a substantially lower Delta distance to the Shakespeare canon than a contemporary English prose control (0.984 vs. 2.148), confirming pronounced stylometric movement toward the historical authorial centroid under the selected MFW configuration.

5 Limitations

We explicitly acknowledge several limitations of the present study:

1. **Evaluation Scale:** While the ingestion pipeline encompasses all 37 plays, the adversarial benchmark was evaluated on 10 targeted probes across representative plays.

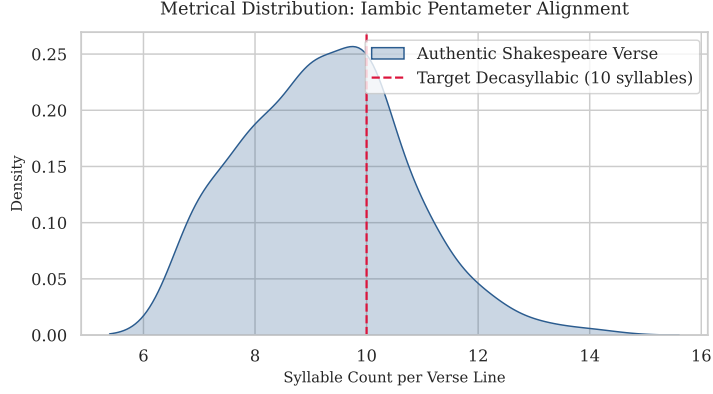


Fig. 2 Baseline kernel density estimation of syllable counts per line across authentic Shakespearean blank verse in the database, demonstrating the decasyllabic reference peak.

2. **Single-Model Scope:** Experiments were conducted using Qwen 2.5-3B; larger models (70B+) or domain-adapted models may display differing baseline leakage behaviors.
3. **Observational Proxy:** Stage presence serves as an operational proxy for knowledge exposure, but does not capture nuanced internal psychological processes such as selective attention, doubt, or intentional deceit.
4. **Phonetic Approximations:** While our CMUdict override lexicon accommodates high-frequency Early Modern contractions, historical pronunciation (Original Pronunciation / OP) varied across the Elizabethan and Jacobean eras.

6 Conclusion

By grounding character retrieval in stage presence and temporal narrative progression, CHARACTERS demonstrates that generative dialogue models can inhabit classic dramatic roles while maintaining dramatic irony. Future work will expand the benchmark suite into a community-annotated Shakespeare Epistemic Evaluation Dataset (SEED) and explore parameter-efficient LoRA fine-tuning for continuous metrical control.

Declarations.

- **Funding:** Not applicable.
- **Conflict of interest:** The author declares no competing interests.
- **Data availability:** The Shakespeare DraCor TEI-XML corpus is open access via <https://dracor.org/shake>.
- **Code availability:** Complete source code for the parser, stage tracker, scansion engine, and Streamlit app is available at <https://github.com/djshaji/characters>.
- **Author contributions:** Sole author responsible for conceptualization, software development, evaluation, and manuscript preparation.

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